

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0312	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
Student Name:	P.Ranjithkumar	Reg. No.:	192411119
Section / Batch:	2024	Date:	06.10.2026

Code of Conduct

I _____P.Ranjithkumar (192411119)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____P.Ranjithkumar_____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____
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Questions

- Q1.** A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.
- Q2.** A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.
- Q3.** A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.
- Q4.** During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log_{10}$ of the ratio).
- Q5.** A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.
- Q6.** An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?
- Q7.** A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?
- Q8.** Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?
- Q9.** A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?
- Q10.** A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

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Co1 - Assessment - Tool - 2

Course Code: ITAB312

Course Name: Mobile Computing Short Answers

Faculty: Dr. Sudha J

1) Full-duplex channels from 45 MHz

A full-duplex channel needs two simplex channels (one uplink, one downlink)

$$\rightarrow \text{Simplex channels} = 45 \text{ MHz} / 25 \text{ kHz} = 45,000 \text{ kHz} / 25 \text{ kHz} = 1,800$$

$$\rightarrow \text{Full-duplex channels} = 1,800 / 2 = 900$$

900 full-duplex channels

2) Channels per cell (7-cell cluster)

$$\text{Channels per cell} = \text{total channels} / \text{cluster size} = 560 / 7$$

80 channels per cell

3) GSM Simultaneous Voice channels

$$\text{Total Channels} = \text{Carriers} \times \text{time slots per carrier} = 100 \times 8$$

800 simultaneous voice channels

800 channels

4) Power Control ratio

$$\text{Ratio in dB} = 10 \log_{10} (P_{BS} / P_{MS}) = 10 \log_{10} (80 / 4) = 10 \log_{10} (20)$$

$$= 10 \times 1.301$$

$$= 13 \text{ dB} \text{ (13.01 dB)}$$

5) Carrier-to-Interference ratio

When both values are in dBm, subtract them

$$C/I = (-60) - (-92) = -60 + 92$$

$$C/I = 32 \text{ dB}$$

A C/I of 32 dB is very good. GSM typically needs about 9 to 12 dB

6) channels per cell (42 MHz, 300 kHz per channel, N=7)

The problem gives 300 kHz per channel so I treat each as a channel directly (no duplexing is stated)

$$\rightarrow \text{Total channels} = 42 \text{ MHz} / 300 \text{ kHz} = 42000 / 300 = 140$$

$$\rightarrow \text{Channels per cell} = 140 / 7 = 20$$

20 channels per cell

7) Received power after doubling the distance (n=4)

Received power follows $P_r \propto 1/d^n$

Doubling the distance changes power by $(1/2)^4 = 1/16$

$$P_r = 64 \text{ W} / 16$$

$$= 4 \text{ W}$$

8) Doubling cell radius

Inverse-square law means:

power $\propto$ distance²

New distance = 2 times

power factor = 2²

$$= 4$$

9) cluster channel count

Given

4 cells

15 channels/cell

$$\text{Total} = 4 \times 15$$

$$= 60$$

10) Doppler shift

Given

$$\text{Speed} = 100 \text{ km/h}$$

$$\text{Frequency} = 900 \text{ MHz}$$

Converted Speed

$$100 \text{ km/h} = 27.78 \text{ m/s}$$

$$\text{Doppler shift} = v f / c = (27.78 \times 900 \times 10^6) / (3 \times 10^8) = 83.34 \text{ Hz}$$